

RAO HOSTEL BOOKING AND MANAGEMENT INFORMATION SYSTEM

Project Proposal and Final Documentation

A PHP and MariaDB Hostel Management Solution

Prepared for academic and project submission

Version 1.0 | 24 August 2026

Part I: Project Proposal

1. Introduction

RAO HB MIS is proposed as a centralized web-based system for managing hostel operations. It will replace fragmented manual records with a controlled application for student registration, room management, bookings, allocations, payments, users, and reports.

2. Problem Statement

Manual hostel administration can lead to duplicated records, delayed room allocation, inaccurate room availability, weak payment tracking, and limited visibility for management. Staff need a single source of current information and clear role-based access to perform daily operations reliably.

3. Aim and Objectives

Aim: To develop a secure and practical hostel booking and management information system.

- Centralize hostel and student records.
- Improve room availability and allocation tracking.
- Manage bookings, payments, and reports consistently.
- Protect information through authentication and user roles.
- Provide AI-assisted operational questions and room suggestions.

4. Scope

Included	Excluded from the current version
Students, rooms, bookings, allocations, payments, users, reports, dashboard, OpenAI assistant, and room recommendations.	Online payment gateway, SMS/email automation, student self-service portal, and automatic AI allocation approval.

5. Proposed Users

Role	Responsibilities
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Administrator	Manage users and all hostel operations.
Warden	Manage students, rooms, bookings, allocations, payments, and reports.
Staff	View and manage permitted operational records and reports.

6. Proposed Technology

- Frontend: HTML, CSS, and JavaScript.
- Backend: PHP with session authentication.
- Database: MySQL/MariaDB using PDO.
- Deployment: XAMPP on Windows for local operation.
- AI integration: OpenAI Responses API through server-side PHP cURL.

7. Expected Benefits

- Faster access to reliable hostel information.
- Better control of room capacity, gender matching, and status.
- Reduced duplication and manual calculation.
- Improved accountability through user roles.
- Faster operational insight through dashboard summaries and AI assistance.

8. Deliverables

1. Working PHP hostel management application.
2. MariaDB schema and initial administrator account.
3. OpenAI assistant and room recommendation endpoints.
4. Setup guide and final project documentation.

9. Implementation Plan

Stage	Work
Analysis	Identify hostel workflows, records, roles, and validation rules.
Design	Define database tables, relationships, pages, and access controls.
Development	Build authentication, CRUD modules, reports, dashboard, and AI endpoints.

Testing	Validate syntax, database relationships, authorization, business rules, and AI failure handling.
Deployment	Install through XAMPP, import the schema, configure OpenAI, and provide documentation.

10. Conclusion of Proposal

The proposed system is feasible using the existing XAMPP environment and provides a focused response to the operational needs of a hostel. The design keeps AI under staff supervision and preserves the database as the authoritative source of records.

Part II: Final Project Documentation

1. Completed System Overview

RAO HB MIS was implemented as a PHP and MariaDB application. It provides authenticated management pages for hostel data and a dashboard showing current operational totals. The completed system also includes two OpenAI-powered features.

2. Completed Modules

Module	Completed functionality
Authentication	Login, logout, password hashing, sessions, and role checks.
Students	Add, view, edit, delete, and validate student records.
Rooms	Manage room number, type, capacity, floor, gender, and status.
Bookings	Create and manage booking records with status validation.
Allocations	Assign confirmed students to rooms while checking capacity and gender.
Payments	Record payment amount, method, reference, date, and status.
Reports	Display student, room, booking, allocation, and payment summaries.
AI tools	Answer hostel questions and suggest rooms for awaiting allocations.

3. Database Design

The database is named `rao_hbmis`. The schema is provided in `database/schema.sql`.

- `users` : user accounts and roles.
- `students` : student profiles.
- `rooms` : room details and availability.
- `bookings` : student booking requests.
- `allocations` : room assignments and dates.
- `payments` : payment transactions and statuses.

Foreign keys protect referenced students and rooms. Unique keys prevent duplicate user emails, student IDs, room numbers, and payment references.

4. AI Implementation

`config/openai.php` sends requests to OpenAI from the server. `api/assistant.php` provides current summary data for staff questions. `api/room_recommendations.php` sends confirmed unallocated students and available rooms for matching suggestions.

Operational control: AI suggestions are display-only. Staff must review and create any booking or allocation through the normal management workflow.

5. Security and Validation

- Protected pages require a logged-in session.
- Role checks restrict administrative actions.
- Passwords use PHP `password_hash()` and `password_verify()`.
- PDO prepared statements are used for database input.
- Database output is HTML-escaped.
- AI request input is limited and API keys remain server-side.

6. Installation

1. Copy the project to `C:\xampp\htdocs\RAO_HBMIS`.
2. Start Apache and MySQL in XAMPP.
3. Open phpMyAdmin at `http://localhost/phpmyadmin`.
4. Import `database/schema.sql`.
5. Open `http://localhost/RAO_HBMIS/auth/login.php`.

Initial login: `admin@rao-hbmis.local` / `Admin@123`. Change this password after the first login.

7. OpenAI Configuration

```
setx OPENAI_API_KEY "your-openai-api-key"  
setx OPENAI_MODEL "gpt-4o-mini"
```

Restart the XAMPP Control Panel and Apache after setting the environment variables. Ensure the PHP cURL extension is enabled in `C:\xampp\php\php.ini`.

8. Testing and Verification

- PHP syntax validation passed for the changed PHP files.
- Editor checks reported no errors in the AI integration files and dashboard.
- The schema matches the application queries and form fields.
- The project was committed and pushed to GitHub on the `main` branch.

9. Limitations and Future Work

- Add CSRF protection to state-changing forms.
- Add audit logging, search, filters, and pagination.
- Add password reset and account lockout functionality.
- Add notifications and downloadable reports.
- Add student self-service access.
- Use HTTPS, a dedicated database user, and production environment configuration.

10. Final Conclusion

RAO HBMIS delivers the planned core hostel management capabilities in a single application. It combines structured database records, role-based workflows, operational reports, and supervised AI assistance. The result is a practical foundation that can be extended for production deployment and additional student services.

11. Project Repository

GitHub: <https://github.com/wambojay-max/R-HBMIS>